

Final Project Proposal: Mitochondrial Diversity in Woolly Mammoths

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Project title: Loss of Mitochondrial Genetic Diversity in Woolly Mammoths Across the Pre-Extinction Bottleneck

Project type: Analysis workflow. An end-to-end pipeline that takes public GenBank mitogenomes to a reproducible, documented statistical result, with a possible Snakemake or Nextflow wrapper (see Sections 3 and 7).

1. Problem statement

We will test whether mitochondrial genetic diversity in woolly mammoths dropped after a small founding group became isolated on Wrangel Island about 10,000 years ago, when rising seas cut the island off from mainland Siberia. Using public, radiocarbon-dated mitogenomes, we compare 28 pre-isolation samples with 14 post-isolation Wrangel Island samples, computing nucleotide diversity (π), haplotype diversity, and segregating sites for each group and testing whether the difference is statistically significant. This matters because small isolated populations are the textbook case in conservation genetics and Wrangel offers a rare before-and-after dataset, but our claim is limited to diversity loss after isolation, not the cause of extinction.

2. Data source(s)

All data come from GenBank (NCBI Nucleotide), and we have already downloaded it: 42 complete woolly mammoth mitogenomes, about 16.9 kb each once aligned. The accession list and radiocarbon dates come from Table 1 of Pečnerová et al. 2017 (*Evolution Letters*), an open-access paper built around this exact pre- and post-isolation comparison. Fourteen samples are post-bottleneck (Wrangel Island, under 10,000 calibrated years BP) and 28 are pre-bottleneck. The new sequences from that study are accessions MG334264–MG334285; the rest were published earlier by other groups and are cited in the same table.

Access is open: we fetch records with NCBI's E-utilities from a plain accession list, with no account, request, or approval needed. The GenBank records carry no use restrictions, and we will cite the original studies. One caveat is that six pre-bottleneck samples have only bounded or missing calibrated dates, so they can be binned but not placed on a precise age axis.

3. Planned methods and tools

We are building an analysis workflow in R, using MAFFT for alignment, `renv` and a conda environment file for reproducibility, and git on GitHub for version control. The first five steps below are done or underway in our public repository (14 commits since Sept 15).

Completed so far

1. **Data acquisition:** built the 42-accession table and downloaded every sequence from GenBank with a scripted command-line pull.
2. **Alignment:** aligned all 42 sequences with MAFFT and verified in R that every sequence has the same length (16,905 bp).
3. **Substitution model:** compared models with `ape` and `phangorn`. AIC and AICc chose HKY+G(4)+I and BIC chose the simpler HKY+I; we kept HKY+G(4)+I because mitochondrial DNA is known to vary in rate across sites.
4. **Tree:** built and plotted a maximum-likelihood tree. The oldest samples (45,000+ years) sit on long branches, and one pre-bottleneck sample (P011), which has many missing sites, falls near the post-bottleneck clade. We kept it as pre-bottleneck based on its radiocarbon date and published haplotype.
5. **Time binning (in progress):** splitting the alignment into the 14 post-bottleneck and 28 pre-bottleneck sequences in R.

Still to build

- **Diversity statistics (our new technique):** π , haplotype diversity, and segregating sites for each group, using `pegas` and `ape` with a manual π calculation as a cross-check. Mitochondrial DNA is haploid, so haplotype diversity is the analogue of heterozygosity here.
- **Hypothesis test (not decided yet):** the candidates are a permutation test that shuffles the pre/post labels, or a regression of diversity against sample age. We will choose after seeing the diversity values and how the six undated samples affect the regression option.

- **Workflow wrapper:** all our scripts are in R. We are open to wrapping the pipeline in Snakemake or Nextflow to meet the workflow requirement; neither of us has used either yet.

Course concepts: Week 6 substitution models; Week 8 multiple sequence alignment; Week 10 phylogenetics and workflow management; the chi-square testing logic from Problem Set 1 (fail to reject versus accept), applied to a permutation test; and git-based reproducibility.

4. Generative AI plan

We use Claude (Anthropic) for specific, checkable tasks and validate every output against the data, the source paper, or an independent calculation. We have used it so far to build the accession table from a paper's table, write the command-line download and MAFFT steps, and draft R code; going forward we will also use it to draft and debug the diversity-statistics and test code, explain statistical choices, and edit the report text.

Documentation. Every AI use goes into `AI_USAGE.md`: the model, what we were doing, our prompt verbatim, the output, how we verified it, and where we corrected or rejected it. We also mark which code is AI-drafted and which is original.

Validation.

- **Data:** the accession CSV was checked against Table 1 of the source paper; sequence counts are checked at each step (42 in, 42 out).
- **Alignment:** all 42 sequences must have equal length, which we check in R.
- **Tree:** clade membership must match the pre/post labels, and any outlier (like P011) must be explained by dates, missing data, or published haplotypes.
- **Diversity statistics:** `pegas` and `ape` results are cross-checked against a manual π calculation, and compared with the haplotype counts reported in the source paper.
- **Hypothesis test:** we run it on shuffled labels first, where no difference should appear, before trusting the real result.

AI-written statistics or code that we cannot reproduce independently are not used.

5. Team roles

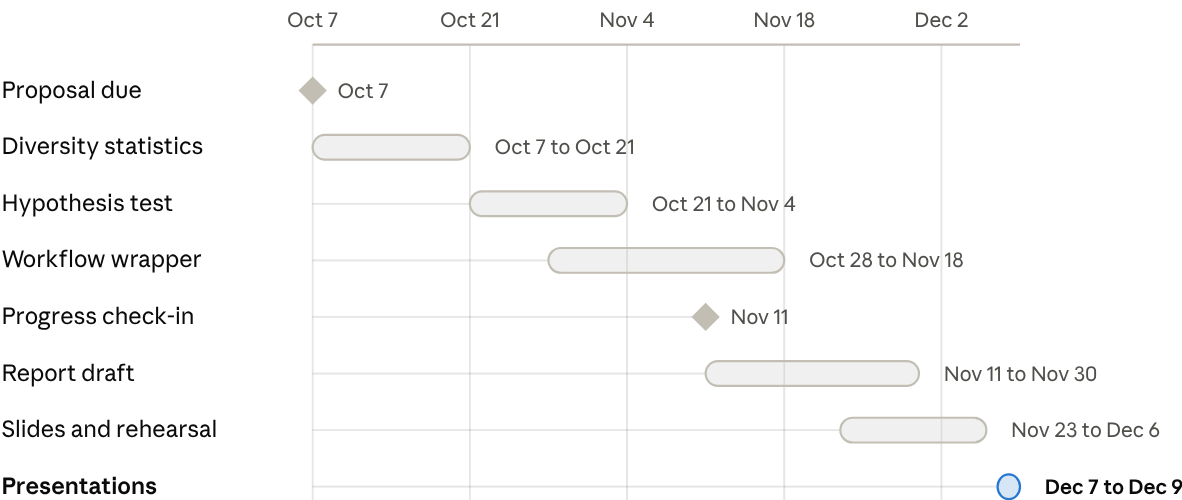
Nicole and Thalie split the remaining work equally across the diversity statistics, the hypothesis test, the workflow wrapper, the report, and the presentation. The preliminary work in Section 3 lives in Nicole's GitHub repository. Both members contribute commits to

the shared repository and entries to the AI usage log, and the report’s Contributions section will list each member’s specific work.

6. Timeline

Analysis is done by Nov 4, leaving a month to write and rehearse

Remaining work, Oct 7 to Dec 9, 2026. Preliminary work is listed in Section 3.



remaining milestones · Oct 7 to Dec 9

Diversity statistics and the hypothesis test come first, then the workflow wrapper (after we have seen the workflow-management material), with the report and slides overlapping the check-in and presentation dates.

7. Risks / open questions

The main risk is statistical power, because 14 versus 28 samples is small and unequal; we will report effect sizes with uncertainty rather than lean on a single p-value. The other risks and our planned responses:

- **Missing data:** P011 and other sequences with many gaps could distort π . We will run the statistics with and without P011 and compare gap-handling options.
- **Binning by date, not place:** two samples found on Wrangel Island (L410 and E470) predate the isolation, so we group them as pre-bottleneck. We will rerun the comparison with them moved to check the result does not depend on that call.

- **Undated samples:** six pre-bottleneck samples lack precise dates. They stay in the two-group comparison but would drop out of a regression on sample age, which is one reason the test choice is still open.
- **Workflow tool:** neither of us has used Snakemake or Nextflow, and our scripts are in R. If we cannot learn one in time, we fall back to a documented script order with environment files, and we would like the instructor's advice on which tool fits best.
- **One locus:** mitochondrial DNA is a single maternally inherited locus, so our result speaks to mitochondrial lineages only, not genome-wide heterozygosity.

Open question for the instructor: if sample sizes allow, is a third intermediate time bin worth adding, or should we stay with two?